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NONINTRUSIVE FLOWFIELD, TEMPERATURE AND SPECIES MEASUREMENTS ON A GENERIC AEROENGINE COMBUSTOR AT ELEVATED PRESSURES

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ABSTRACT

A generic combustor was built, that gives wide optical access at higher pressure and shares typical features with aero engine combustors. A comprehensive data set for validation of RANS and LES codes was generated at isothermal as well as combusting conditions at 2 and 10 bars with 650 K preheat using natural gas as fuel. The velocity field was measured using LDA (Laser Doppler Anemometry) and DGV (Doppler Global Velocimetry) as well as PIV (Particle Image Velocimetry). Temperature data were acquired using CARS (Coherent Anti stokes Raman Scattering) and SRS (Spontaneous Raman Scattering). Major species concentrations as well as the mixture fraction in the primary zone of the combustor were also measured using SRS. Mean and RMS values of the temperature measured by CARS in the secondary zone illustrate the influence of the jet impingement on the unsteady mixing of the jets with the swirling primary air. [*Keywords:* non intrusive measurements, aero engine combustion, elevated pressure]

INTRODUCTION

Combustion CFD codes currently used in the design of combustors are based on Reynolds averaged Navier-Stokes (RANS) equations, which are based on a steady-state solution of time-averaged flow equations. Although these codes have proved valuable in the design process, the RANS approach has well known deficiencies in accurately predicting features of turbulent flows like the spreading and mixing of turbulent jets, highly turbulent and genuinely unsteady flows. A more accurate prediction of the detailed flow field and distributions of temperature and chemical species will result in higher accuracy predictions of NO_x and other harmful emissions. Some of the limitations of RANS codes can be overcome by the Large Eddy Simulation (LES) method, where time-dependent turbulent fluctuations of a size larger than the size of the numerical grid are resolved accurately in time. Within the MOLECULES project of the 5th EU Framework Program Growth [1], validated LES codes capable of simulating the turbulent, unsteady flow inside a combustor were to be provided to the aero engine industry with accurate, high-quality databases of validation data to compare the performance of the LES- against the current RANS codes in configurations sharing critical features with realistic combustors.

Confidence in CFD as a design tool cannot be solely built on validation experiments that represent one aspect of the flow where the corresponding physical model is tested in isolation. Only a test case presenting a rather comprehensive range of effects found inside the combustion chamber in relevant geometries and at higher pressure will enable to judge the ability of the codes to optimally perform a practical calculation. Since the complexity and operational cost of a real aero engine sector combustor prevents the application of measurement techniques to fully catch the range of phenomena [2], compromises had to be made and a generic combustor was selected. It should nevertheless represent the conventional combustor architecture with a diffusion burning primary zone, a mixing zone with colliding jets and film cooled liner walls.

A single sector with quadratic cross section was chosen. For the verification of combustion models, the acquisition of mixture fraction data in the primary zone was judged to be essential and thus Spontaneous Raman Scattering was selected as the diagnostic method for temperature and species measurements in the primary zone. Since the Raman signal is too weak to be used in spray combustion, the burner had to use gaseous fuel. Hence a burner for combustion of natural gas with the same aerodynamic features as a generic diffusion burner for spray combustion was developed. For the same reason, sooting flames could not be allowed, which forced the choice of an overall lean air to fuel ratio for the primary zone. Given the possibility to acquire mixture fraction and temperature data, the effect of pressure on reaction progress becomes accessible.

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Consequently two operating conditions, one near atmospheric pressure and one at sufficiently high pressure to observe a difference in kinetic effects, i.e. at 10 bars were chosen. To retain similarity to the flame stabilization in practical combustors, the combustor air was preheated to 650K.

In the remainder of the paper, the specific features of the combustor connected to the optical access are pointed out, the trade off in the burner design between realism and good definition of boundary conditions is explained, the adaptation of the measurement techniques to the experiment is briefly discussed and exemplary results on boundary conditions, primary and secondary zone are highlighted.

EXPERIMENT

Combustor

The Single Sector Combustor (SSC) test bed was used for the experiments [3]. However as hitherto the interest had been focused on the primary zone, its combustor had no optical access to the mixing zone and a new one had to be built. A longitudinal cross section of it is shown in Fig. 1.

The pressure casing has windows at three sides. Two long windows are placed at the side of the sector stretching over primary- and mixing zone and a smaller one, seen in Fig. 1, giving access to the primary zone. In addition there are two windows in the exit plate to be used by Laser light sheets extending into the mixing zone. The casing is cooled with air at normal temperature flowing along the secondary channel around the combustor liner to the longitudinal sonic throat in the exit. It is also used for convective cooling of the liner, film cooling of secondary zone liner windows, the metallic parts of the liner and finally the control of the combustor pressure. Electrically preheated air enters the combustor liner through the burner, the filmer lips for the primary zone window cooling and the mixing jet holes. These have their own secondary channel, which is connected to two tubes leaving the combustor casing. The flow split between these three branches is set by exchangeable sonic nozzles in each flow path. The heated and unheated air flows are independently controlled. The liner cooling flow is monitored by a differential pressure measurement between the inside of the liner and secondary channel. The pressure oscillation inside the liner is measured with a water cooled Kistler sensor mounted flush to the liner wall.

The design of the liner had to be adjusted to the demands of optical access to the primary and secondary zone. The jet mixing was to present a demanding task for the CFD benchmarking; hence a configuration with colliding jets was chosen. The multisector layout with one pair of opposite holes on the burner axis and one on the sector boundary was modified to one central pair of big holes and two pairs of small holes with an area equal to the central pair, resulting in one colliding and two non colliding pairs of jets. The correlation of Holdeman et al. [4] was used to size the holes for equal pressure loss of burner and liner holes. For the isothermal operation, the mass flow was changed using different critical nozzles to preserve the same momentum ratio for primary and mixing air as in the hot case.

The optical access to mixing jets had to allow imaging of the downstream region but also to the upstream area where the

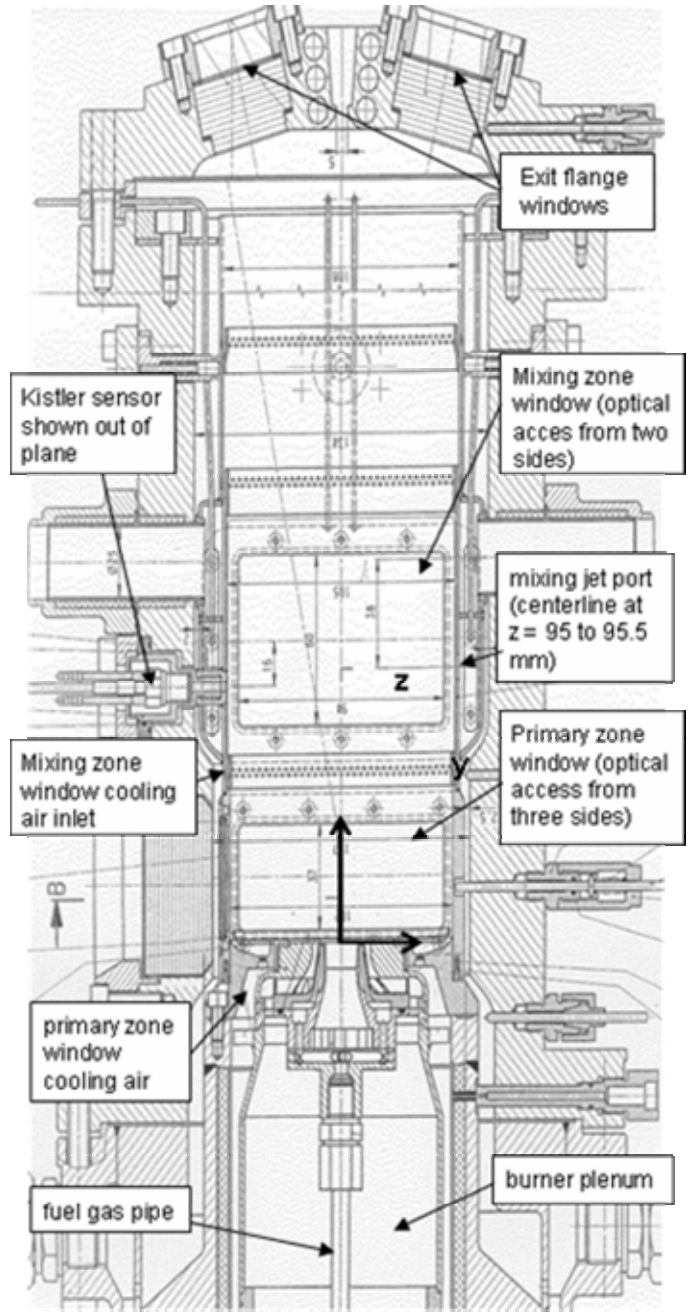

Figure 1: Cross sectional view of single sector combustor

mass from the colliding jets would penetrate. For the resulting 60 mm window length, film cooling with ambient temperature air was judged necessary. The length of the primary zone was chosen to result in a nearly cubic volume. With the minimum length needed for the rims of the liner windows and the window cooling holes, the length of the primary zone window is restricted to 37 mm.

Burner

The calculation of the swirling, recirculating primary zone flow depends to a wide extent on the accurate formulation of its inflow boundary condition. Hence the use of practical hardware was foregone in favor of a burner designed to allow the best

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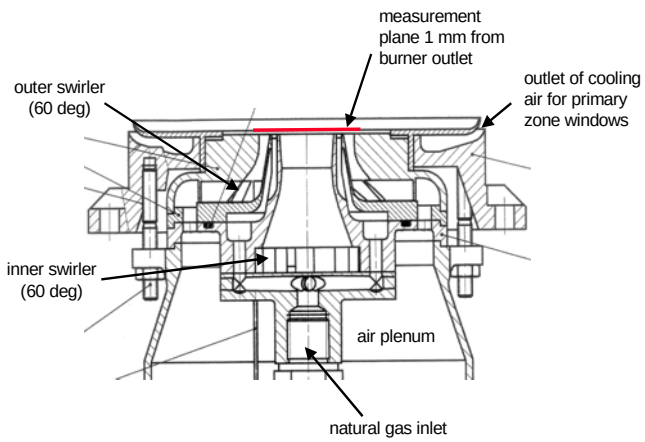

Figure 2: Longitudinal section of the burner

possible measurement of the flow at the burner outlet. Nevertheless it should emulate a kerosene diffusion burner while using natural gas as fuel. A preexisting design of a gas film nozzle for atmospheric rigs was adapted to the SSC, see Fig. 2. The basic idea is to replace the fuel film inside the swirl cup of a double swirl burner with a co-annular fuel gas flow of about the same momentum. In contrast to industrial burners with fuel gas injection, the momentum of the air flow should dominate over that of the fuel gas flow and the mixing of the evaporating liquid fuel is mimicked by the entrainment of the lower velocity circular fuel jet.

For the even distribution of the gaseous fuel around the circumference, the co annular channel was replaced by an array of 36 rectangular slots of 1.5 x 1 mm with solid bars in between, which ended 1 mm before the outlet of the burner. For an easy definition of burner outlet conditions, the recirculation of combustor air in the burner should be minimized and the burner flow should be measured as early as possible. Hence it was decided to let the swirl channels end in a straight section at the combustor inlet plane and to relinquish the diffuser which is normally used to augment the recirculated mass for a given amount of swirl [5]. To make up for that effect and to reach a swirl number for the gaseous flow including the nonswirling fuel gas flow, that guarantees stable recirculation and flame position, co swirl and a relatively high swirl angle of 60° was chosen and radial swirlers were used for ease of manufacturing.

The resulting flame is shown in Fig. 3 at 4 bar and 700K with a view through the primary and secondary liner windows. Even with the high swirl, the expansion of the swirling flow is moderate. The distance needed to entrain the fuel gas jets represents an analog to the atomization, vaporization and mixing distance found in spray burners. Hence it is bound to exhibit a similar mixture of diffusion burning with partial premixing as spray burners with good atomization. However because of the long axial extension of the heat release zone resulting from that, the latter part of that zone is hidden by the rim between primary and secondary zone window.

Operating conditions

Table 1 gives an overview of the operating conditions. Since the control of the combustor pressure is effected with the

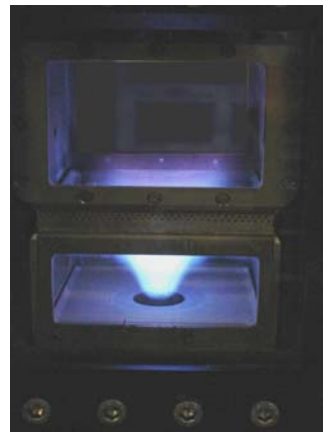

Figure 3: Photograph of the flame at a pressure of 4 bar and preheat temperature of 700 K

critical nozzle at the outlet and the cooling air mass flow, the lower pressure operating condition was not atmospheric but at 2 bar. All tests were done at a pressure loss of 2.75%, which gave enough margin to the onset of an instability developing at higher pressure losses. As previously mentioned, the isothermal test was done with an equivalent volumetric air flow for the fuel gas and a higher ratio of primary to secondary air mass flow to compensate for the nonexistent temperature difference in the expression for the jet momentum ratio.

Operating condition	Cold	Hot	hot
combustor pressure	2 bar	2 bar	10 bar
burner pressure drop	2.75%	2.75%	2.75%
plenum temperature	280 K	650 K	650 K
secondary air temp.	280 K	670 K	680 K
AFR	no fuel	25.5	25.5
burner pressure drop	0.055bar	0.055bar	0.275bar
burner air mass flow	52.3 g/s	35.7 g/s	181 g/s
natural gas mass flow	3.4 g/s air	1.4 g/s	7.1 g/s
prim. zone film cooling air	11.2 g/s	5.9 g/s	26.4 g/s
mixing air mass flow	18.0 g/s	22.3 g/s	112 g/s
mixing zone film cooling	28 g/s	12 g/s	48 g/s

Table 1: Operating conditions of Single Sector Combustor

upper calorific value	10.19	kWh/m³
lower calorific value	9.191	kWh/m³[1]
density at 1.013 bar, 0°C	0.7939	kg/m³
relative density[2]	0.614	-
Wobbe index	13.005	kWh/m³

Table 2: Combustion related physico-chemical properties of natural gas

Because of the appreciable fuel gas flows at higher pressure, the in house natural gas supply facilities were used which deliver a so called Low Gas with higher nitrogen

[1] equivalent to 40861 kJ/kg
[2] relative to air

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content. The chemical composition and the combustion related physico-chemical properties are given in table 2 and 3.

Species	Mol-%
Carbon dioxide CO2	0.8385
Nitrogen N2	9.1064
Oxygen O2+Argon Ar	< 0.01
Methane CH4	87.874
Ethane C2H6	1.7695
Propane C3H8	0.3083
n-Butane C4H10	0.0426
i-Butane C4H10	0.0303
n-Pentane C5H12	0.0057
i-Pentane C5H12	0.0067
Neo-Pentane C5H12	0.0019
C6 and higher	0.0161

Table 3: Chemical composition of the used natural gas

MEASUREMENT TECHNIQUES

Flow velocities were measured by LDA, PIV and DGV. The burner outflow and three profiles from 10 to 35 mm from the burner as well as several profiles around the mixing jets were measured with LDA. Dantec 2-D sending optics and a fast in house receiving optic operating at a 13° off axis angle were used. The signals were processed by Dantec Burst Spectrum Analyzers. The resulting dimensions of the measuring volume were a diameter of 150 µm and a length of 1.5 mm. The airflow was seeded with liquid particles of a glycerin and ethanol mixture for the isothermal- and with Al_2O_3 for the combusting case, both with diameters below 1µm. Tests with a calibrated, contoured nozzle gave an agreement of better than 1%. For the measurement at the burner outlet, the optics were tilted by 9° to circumvent the filmer lip for the window cooling air, see Fig. 2. The resulting statistical moments can be transformed to the original coordinate system. Spectral measurements were performed at the outlet and a height of 10 mm for isothermal and combusting flow. The data processing is described in [6].

The velocity field of the primary zone was measured by DGV. Its application to combusting flows is described in [7]. Only mean velocities are measured with the used setup with on chip integration of signals. On the other hand a very dense volumetric data set was produced, mapping out the primary zone from z = 2 to 34 mm in 2 mm steps and in 1 mm steps for the two other directions for isothermal and combusting flow. The detection limit of the technique is 1 m/s. As it relies on the measurement of intensity differences, special care has to be taken to avoid reflections and secondary scattering. A dynamical background signal correction and turbulence correction were introduced as new techniques to the data processing. However preference must be given to LDA data at the places where they are available.

PIV was used to measure the velocities in the secondary zone. For the isothermal measurement, a Stereo-PIV setup could be used, that gives instantaneous 3-D velocity maps. Initial conditions of the jets were acquired at 1 mm distance from the holes. For the combusting experiment, only 2-d PIV was possible. A combination of 2-D PIV and DGV for the third velocity component was realized for this experiment to attain 3D velocity data without 3-way optical axis in the mixing zone. The windows in the outlet were used for the Laser light sheet and the side windows for the imaging. The single shot uncertainty was calculated to 2-4 m/s, the resulting uncertainty for the mean velocity with 1000 shots to 0.1-0.2 m/s. The precise optical arrangement can be found in [8].

In the primary zone, laser Raman scattering was used for the simultaneous determination of the major species concentrations (CO_2, O_2, CO, N_2, CH_4, H_2O, and H2), mixture fraction, and temperature. At a total of 1800 measurement locations on 5 heights (z = 10, 24, 18, 22, 26 mm) 1000 single shots were taken. A critical point was the irradiation of the focused high-power laser beam through the combustion chamber windows. The requirements for high signal levels and good spatial resolution in a 1D configuration were inevitably connected with the risk of window damage. Therefore a pulse train from 3 double-pulse Nd: YAG lasers at λ=532 nm was used for the excitation. In order to reduce the intensity at the windows and to avoid an optical breakdown in the focal region, the pulse train of the six laser pulses is elongated by a pulse stretcher, and furthermore the laser beams were focused by a combination of two cylindrical 250 mm focal-length quartz lenses to generate a “blurred” focus of an average diameter of 0.5 mm. In this way, the total pulse energy of 1 J could be irradiated into the combustor. The signal levels generated with this setup were sufficiently high to allow for single-shot measurements. The optical setup, the calibration and evaluation procedure of the signals are described in [9].

The temperatures in the secondary zone were measured by a commercial broadband CARS system (SOPRA). Data evaluation and part of the data acquisition were developed at DLR and especially adapted for measurements at high temperature and pressure [10]. The experiments were performed in a planar BOX-CARS beam arrangement on nitrogen. An achromatic lens with a focal length of 300 mm was used to focus the laser beams. The corresponding probe volume, in which 95% of the CARS signal is generated, had a length of 3.2 mm under laboratory conditions.

Due to the broadband operation of the dye laser with each laser pulse it is possible to record a complete nitrogen CARS spectrum, which can be evaluated giving a single pulse temperature. A typical measurement ensemble (temperature PDF) contains between a few hundred and 1200 single pulse temperatures. From these ensembles temperature mean value, rms (T) and some other statistical information are extracted. At 2 and 10 bars 40 measurements were taken in each of the 3 y-z planes at x=-10, 0 10 mm around the main jets on 5 heights at z = 85, 95, 105, 115 and 125 mm between -35 < y < 35 mm.

Prior to the experiments the status of the CARS system and the data evaluation strategy were checked by temperature measurements on an atmospheric oven cell with a homogeneous and time stable temperature profile. The measured mean temperature of 1560 K agreed better than 1 % with data from a newly calibrated two colour pyrometer. For laboratory burner measurements the temperature accuracy under atmospheric conditions is about 2 % for the mean temperature and 2-3 % for the root mean square value rms(T), and at 10 bar about 3 % for the mean temperature and 4-5 % for rms(T), reflecting the lower temperature sensitivity of the CARS spectra.

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RESULTS

Boundary conditions

LDA measurements were performed 1 mm from the burner outlet at all operating conditions. The mean velocity profiles of the isothermal condition are given in Fig. 4.

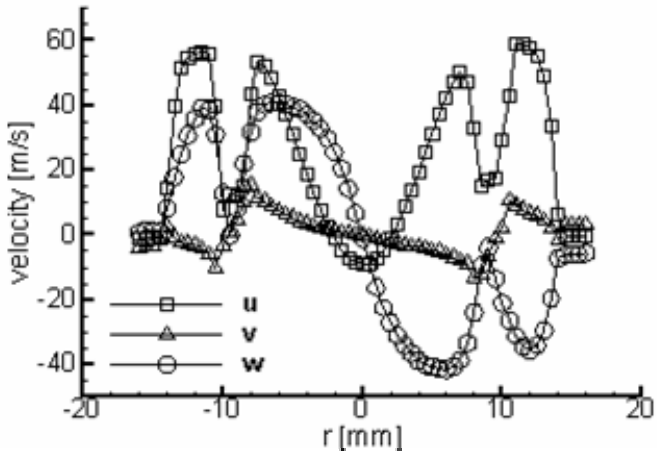

Figure 4: Profile of axial (u), radial (v) and tangential (w) velocity 1 mm from burner outlet at isothermal condition

The assumption of axial symmetry is shown to be justified. The wake of the fuel ring at r = 10 mm is clearly seen in the axial- and tangential profile, with a nearly zero value of the tangential velocity at the center of the slot. The axial velocity over a short part of the fuel ring is presented in Fig. 5. The remnants of the discrete fuel jets are still noticeable at this position 2 mm from the exit of the fuel ring, c.f. Fig. 2.

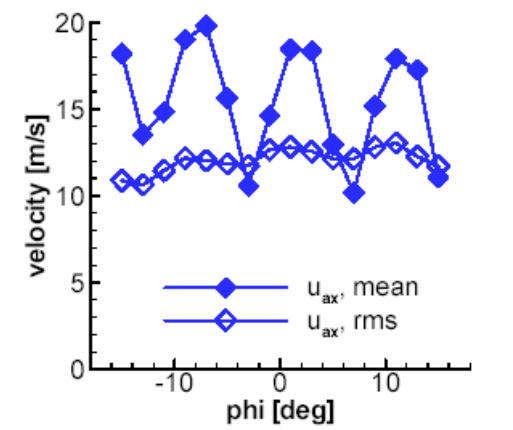

Figure 5: Axial velocity over fuel ring at z = 1 mm at isothermal condition

The profiles of the RMS velocity largely follow the gradient hypothesis; however the level is higher in the inner swirl stream, where all components reach a peak value of 25 m/s. Furthermore the peak of the radial RMS velocity is at r = 0 mm, which suggests an oscillation of the axis of the flow. Therefore, spectral analysis of the LDA data was performed.

The spectra of the data obtained in the core of the burner flow exhibit a pronounced peak near 1.7 KHz for both the radial as well as the tangential component. No preferred frequencies were found for the autocorrelation of the axial component nor for any cross- correlation involving axial velocity. The flow of the burner annulus (r = 12 mm) exhibits no preferred frequencies whatsoever.

The velocity measurement at the jet mixing holes in the cold flow was performed using a Stereo PIV setup permitting simultaneous measurement of all three velocity components. The thickness of the light sheet was 1 mm. Figure 6 shows the mean velocities of the first measurement plane at 1 mm from the central hole, the velocity components in the plane are presented as vectors while the component normal to the plane is indicated by color contours.

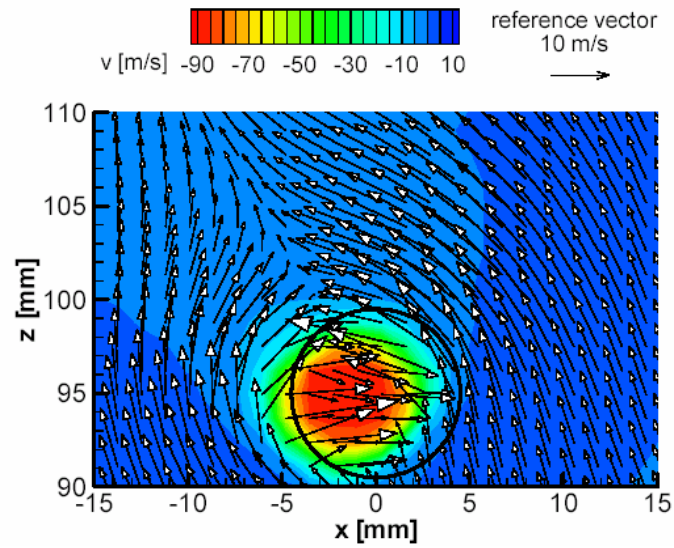

Figure 6: Mean velocities 1 mm from center of mixing jet port at y = 49 mm, isothermal condition, the circle denotes the jet hole contour

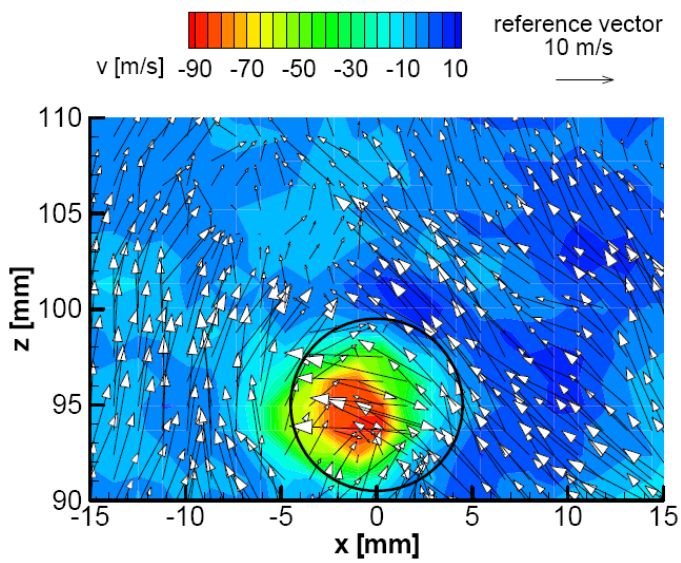

Figure 7: Instantaneous velocities at the same location and condition as Fig. above

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The core of the jet is seen to be smaller as the hole diameter. This is caused by the flow characteristics of their secondary channel, c.f. Fig.1, which had to be rather thin to fit into the pressure casing. Also, the center of the jet is displaced a little to the negative x direction. High in plane velocities are seen at the border of the jet pointing inward to the jet core, reflecting the strong entrainment at the root of the jet. The stagnation line between the primary zone fluids passing to the left and right of the jet is turned by the swirl to an 11 o'clock direction seen from the center of the hole.

In Fig. 7 the instantaneous velocity field is shown for the same location and condition as for Fig. 6. At this instant in time, the high velocity core of the jet has a much smaller extension than in the mean velocity field resulting from the vigorous interaction of primary and jet flow already at the root of the jet. This underlines the necessity to include a sufficient area upstream of the hole in the calculation domain. Furthermore the clear stagnation line in Fig. 7 presents itself at this time increment more as a stagnation region with low in plane velocities around the point x= -3mm, z=103 mm.

Primary Zone

Only results from the 10 bar operating condition with combustion are shown. First, an overview of the burner flow is given with the axial velocities in the region accessible through the primary zone window.

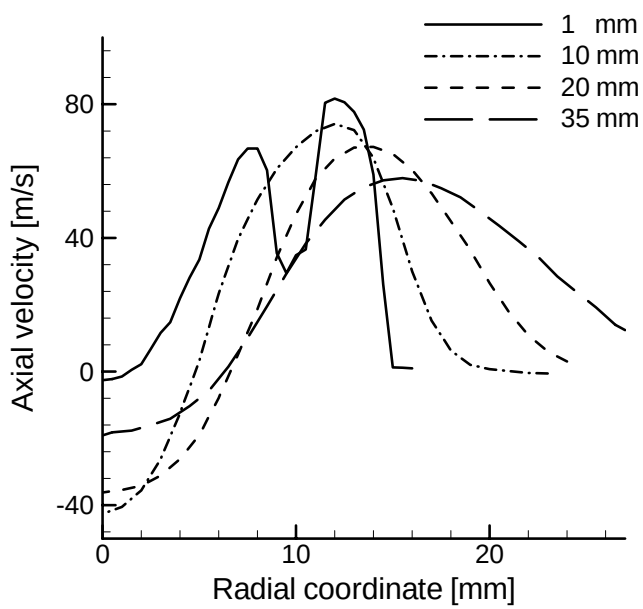

Figure 8: Axial velocities at 1, 10, 20 and 35 mm from the burner for the 10 bar condition

The slow radial expansion of the swirling flow is consistent with the expectation from the photograph of the flame in Fig. 3. From z = 20 mm to z = 35 mm the width of the recirculation zone already decreases and looses in strength such that the recirculated mass remains small. The wake of the fuel ring vanishes at 10 mm whereas it can be clearly seen at that position for the isothermal flow, which had a dummy fuel flow of air with the same volumetric rate but higher mass.

The correspondent RMS velocities are shown in Fig. 9. The small peaks between r = 8 and r = 10 for the first profile show the extra shear generated by the discrete fuel jets. Since the flow is not circumferentially symmetric at this point, cf. Fig. 5, a measurement taken at the other axis has different

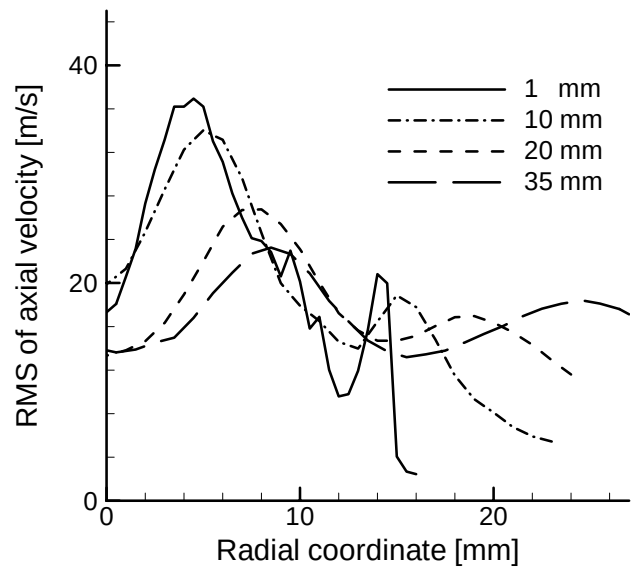

Figure 9: Axial RMS velocities at 1, 10, 20 and 35 mm from the burner for the 10 bar condition

values at this radius. Again the radial RMS velocity near the burner has a peak on the axis and consequently the spectral measurements performed near the burner also show peaks, this time in all velocity components and at 2.5 KHz which gives the same Strouhal number as for the isothermal case. Comparison of hot and cold spectral measurements at z = 10 mm reveals that the peak is progressively damped with the temperature rise from 10 mm > r > 0 mm.

A comparison between velocity and temperature derived from the Raman data at z = 10 mm and z = 26 mm is shown in Fig. 10.

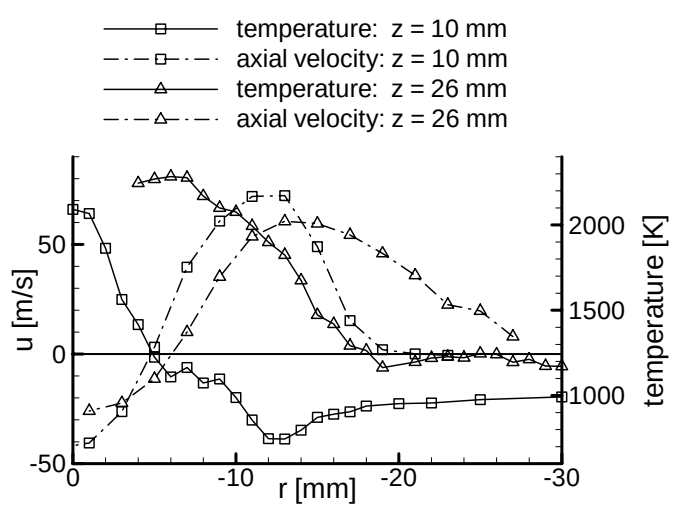

Figure 10: Mean temperatures and axial velocity for the 10 bar condition

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The two chosen heights show the situation before and after ignition of the gaseous fuel stream. At z = 10 mm only the recirculation zone is hot. However, the mixture is within the approximate boundaries of the flammability limit only between 8 mm < r < 13 mm, where the temperature is still rather low and hasn't reached the spontaneous ignition temperature at the stoichiometric point at r = 12 mm. This condition is fulfilled for the mean flow between 10 mm < z < 14 mm. At z = 26 mm, the high temperature region has reached the core of the swirling jet flow. A more detailed view on reaction progress can be gained with a scatter plot of temperatures against mixture fraction. This is done at z = 18 mm in Fig. 11. for three different radii, r = 0, 12 and 26 mm.

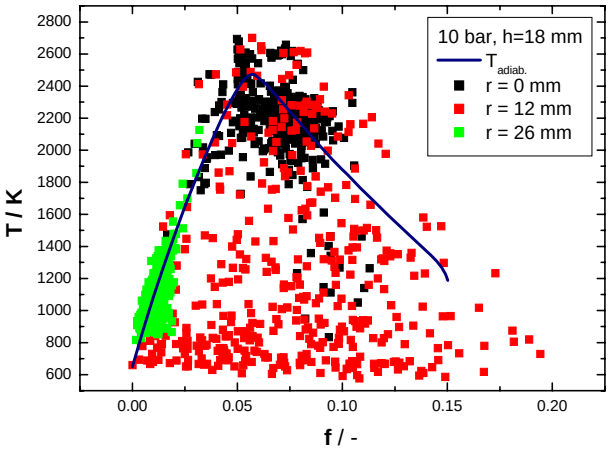

Figure 11: Scatter plot of temperature at 10 bars and z=18 mm. Symbols represent results from single pulse Raman measurements at three radial locations

Not surprisingly, the mean mixture at r = 0 mm is rich with CH_4 and O_2 concentrations below 2% and the temperatures of the burned out mixture high. On the other side at r = 26 mm, the mixture fractions are very lean, the spread in temperature being produced by hot gas mixed in from the lean burning outer recirculation. The CO_2 concentration at this plane reaches a minimum at r = 18 mm, with a slight increase at higher radii marking the occurrence of recirculated gases. At r = 12 mm, the mean mixture fraction is still rich, however the RMS value is at its peak here and hence a wide distribution of points along the abscissa is seen. The temperature distribution seems to be slightly bimodal with a lot of points below 900 K that haven't reacted at all, quite a number of points above 2000 K that are at the approximate end of reaction and a somewhat smaller number of points between those temperatures. Fast chemistry would produce a bimodal distribution with almost no points between the low temperature limit and the adiabatic line.

In summary, the limited amount of recirculated heat and the slow radial expansion of the swirling flow produced a reaction zone with an unusually long axial extension compared to most practical gas turbine burners. On the other hand that leads to a higher axial stratification of heat release and therefore lower axial gradients of species and temperature giving rise to a good spatial resolution of reaction progress in the measurements, which should make comparisons to computed results easier.

Mixing Zone

The mean mixing zone flow is portrayed by two cross sections in the plane of the main jets perpendicular to the z and the x axis. Because of the better quality of the combined PIV and DGV as well as the CARS measurements at the lower pressure, results from the 2 bar condition are selected. Figure 12 shows the velocities in the plane of the jet holes at z=94mm.

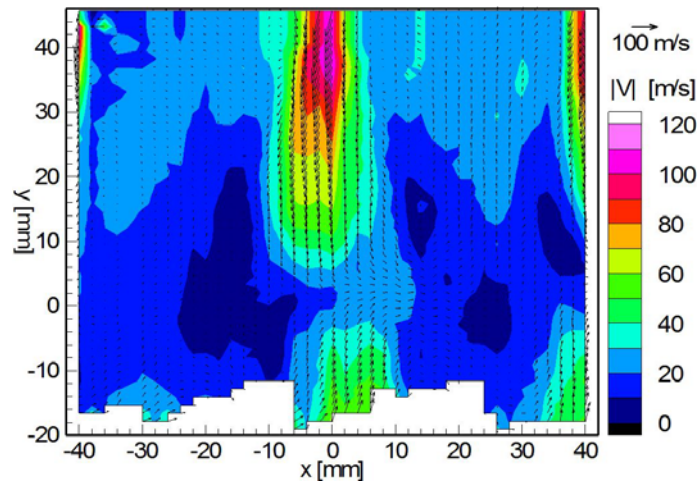

Figure 12: Mean velocities in the plane of the jets at z= 94 mm at 2 bar

The extension of the measuring zone is higher in the positive y direction than in the negative one, because the light sheets were introduced from the exit flange window at the positive y side. The plot is an interpolation from the measurements made in 2 mm increments with light sheets along the y direction. Consequently the velocity component in the x direction results from the DGV measurement and the velocity component in the y direction from the PIV measurement. The big jets are seen to collide, but also to be deflected into the counterclockwise direction under the influence of the swirl.

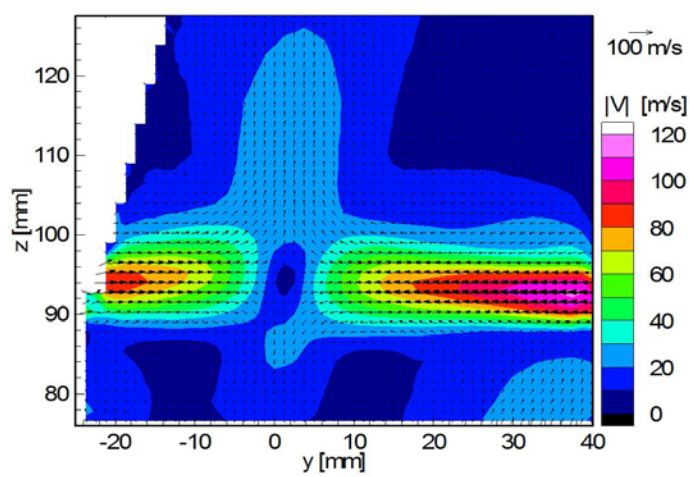

Figure 13: Mean velocities in the plane of the jets at x= 0 mm at 2 bar

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The orthogonal view through the main jets is depicted in Fig. 13. Here the velocities result from PIV measurements of a single light sheet and therefore there is no interpolation. Apart from the slight shift of the stagnation point between the jets to y=2mm, the mean flow is quite symmetric. The jet fluid is mostly deflected to the downstream region, but there is also a non negligible amount recirculated into the primary zone.

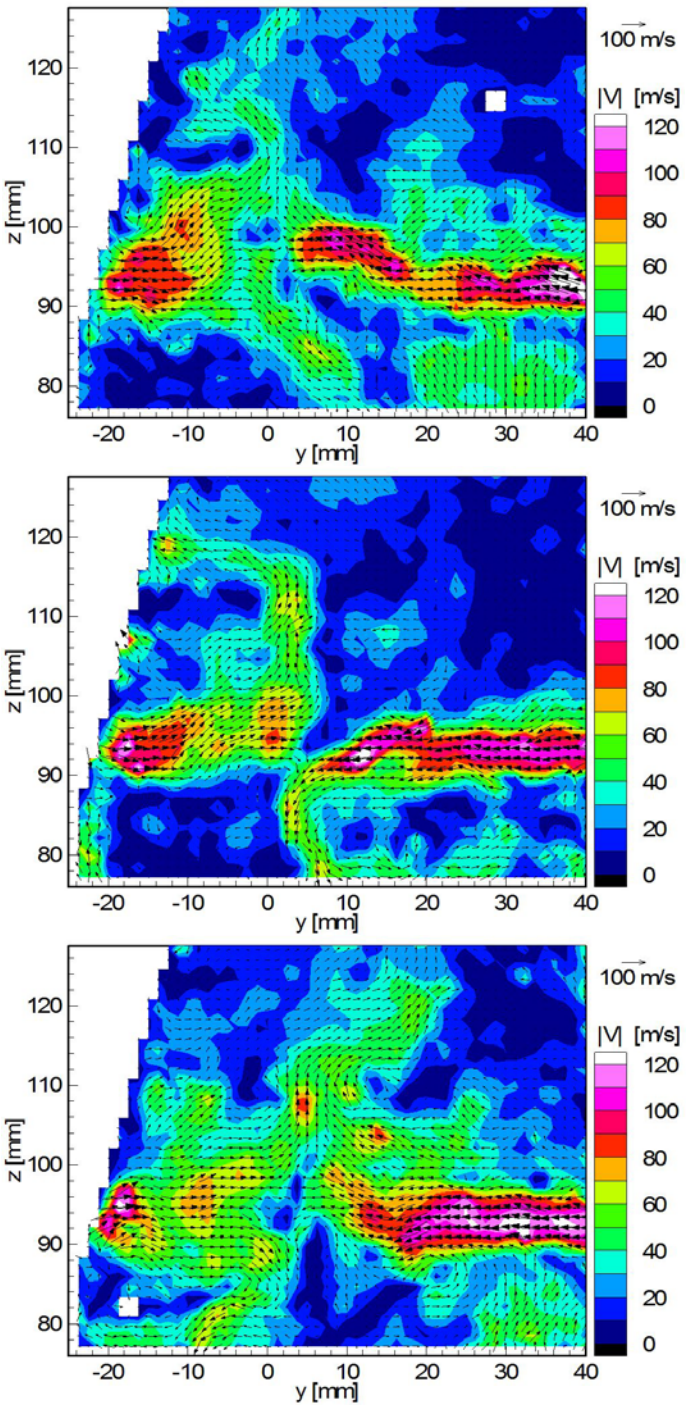

Figure 14: Instantaneous velocities in the plane of the jets at x = 0 mm at 2 bar

Looking at three instantaneous snapshots of the velocities in the same plane depicted in Fig. 14, a different impression can be gained. The main jets collide and burst into rather sharp jet like structures downstream and upstream, that move with high speed in any angle. In the three snapshots we can distinguish a situation where the jet pointing upstream is fed from both main jets, from the jet on the positive y direction on the right and finally from the jet on the negative y side at the left side. Although the mean flow in Fig. 12 is symmetric in the y-z plane and the recirculating flow exhibits smooth gradients, it is genuinely unsteady. Not surprisingly the comparison between RANS and LES calculations that has been performed so far, c.f [11] shows a clear advantage for the LES calculation with the RANS result underpredicting the penetration and overpredicting the spreading angle of the jets.

Although the best complement to the PIV measurements would have been planar temperature measurements, the method using OH-LIF could not be applied because of the lack of OH in the low temperature regions of the jet. Even for the 1-D Raman measurements, the optical access using the windows in the exit flange proved to be insufficient, hence CARS as a reliable single point technique with comparatively high single shot accuracy was chosen. Fig. 1 shows the mean temperature distribution in the same plane as Figs. 13 and 14 for the 2 bar condition.

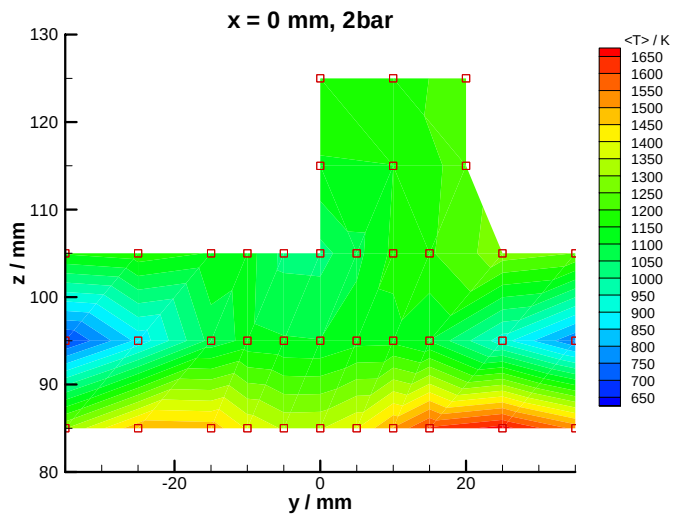

Figure 15: Mean temperatures in the plane of the jets at x = 0 mm at 2 bar

The measurement points are marked with quadrangles. For ease of comparison with the preceding pictures, interpolation was used to arrive at the contour plot in the figure. The cold jets and their mixing with the hot primary air can be easily followed along the z = 95 mm line. When comparing with Fig 13, the smaller measuring range in the y direction has to be kept in mind. The measurement upstream of the jet axis at z = 85 mm shows the end of the primary zone, but also between -5 mm< y < 5 mm a colder region where the cold fluid from the colliding jet is encountered.

Accordingly the map of the RMS temperatures in Fig. 16 shows very high temperature fluctuations where hot entrained primary fluid reaches the centerline of the jets. Although the velocity distributions below the collision at x = 0, z = 85 mm are very

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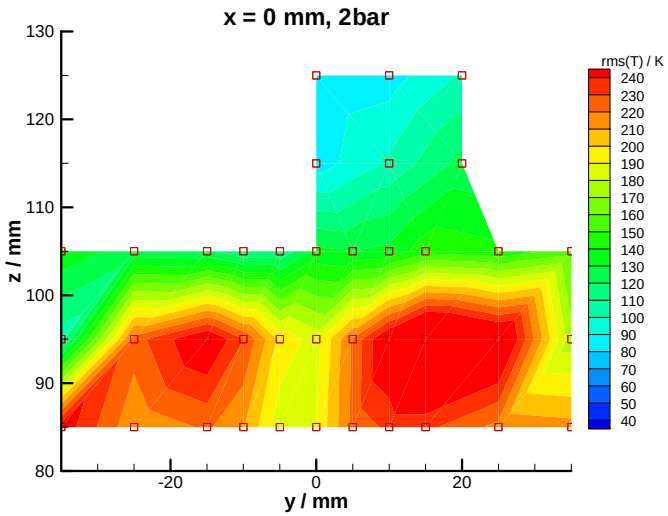

Figure 16: RMS temperatures in the plane of the jets at x = 0 mm at 2 bar

unsteady, with fluid from the jet alternating with fluid of very low velocity, the RMS of the temperature fluctuations are lower than on the centerline of the jets before the point of collision. Here, the temperature PDF of Fig. 17 available from the CARS measurement shows a lack of temperatures between 1800 and 2200 K, which are found elsewhere in the hot zones with higher RMS temperatures. Even the low velocity fluid is already at lower temperature than the primary zone gas, whereas on the jet centerline pure jet fluid coexists with pure primary zone flow and mixtures thereof. Hence for the prediction of the mixing of the secondary air it is also necessary to accurately describe the turbulent mixing of temperature and the turbulence of the primary zone flow.

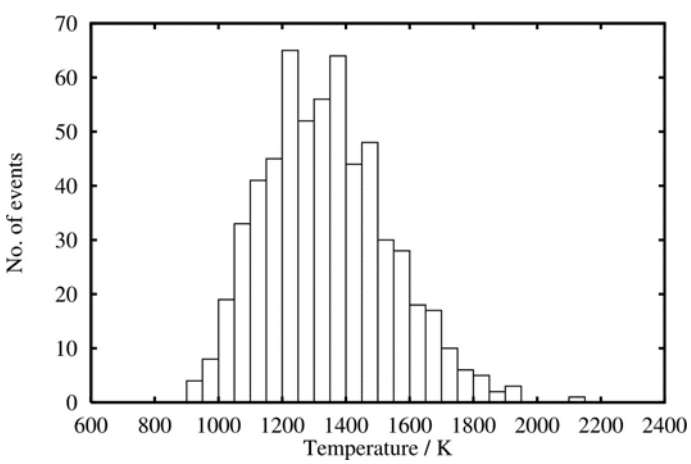

Figure 17: Temperature PDF of CARS measurement at X = Y = 0 mm, z = 85 mm at 2 bar

CONCLUSIONS

The intention of the project to compile a comprehensive database of velocity, temperature and species for the validation of RANS and LES codes has been fulfilled. A dedicated combustor has been built and efforts to adapt measuring techniques have been undertaken to reach a threefold aim:

The reactive flow of the combustor should be representative of a practical aeroengine combustor for the most important aspects of flow and combustion modeling.

The boundary conditions should be described accurately enough to make the calculations as independent as possible from arbitrary choices made for the input to the computation.

The comprehensiveness, the detail and accuracy of the data should be high enough to present a challenging task to the different groups of CFD modelers and users to which the dataset is transferred.

The material presented in this contribution has been selected to enable the readers to judge how far this promise has been fulfilled. An exhaustive description of the physics contained in the dataset was not possible. Instead an overview of the whole combustor flow was given and some exemplary features were highlighted. The interaction of reaction and turbulent mixing in the primary zone could only be described superficially and the marked influence of the operating pressure was not exposed. This will be done in more detail in a subsequent publication. It was however shown that the heat release is neither dominated by mixing or chemical reaction times. With a fuel consisting predominantly of Methane for which good kinetic data is available, it should be a useful test case for combustion models. In the mixing zone, the 3-D field of instantaneous velocity data and the temperature information should constitute a demanding test case for the modeling of the turbulent transport of heat and momentum.

It is anticipated that with more results from the calculations becoming available, the dataset will have to be interrogated and evaluated in many ways to elucidate the differences found between measurement and calculation. For this reason it will be a continuing activity for some time to come.

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